

DEVOPS ASSIGNMENT - Git and GitHub

Question 1

1. Create a new folder
2. Create files: code.txt, log.txt, output.txt
3. Stage code.txt and output.txt
4. Commit them
5. Push to GitHub

Solution - Commands Used

mkdir project1

cd project1

git init

touch code.txt log.txt output.txt

git add code.txt output.txt

git commit -m "Added code and output files"

git remote add origin <your-github-repo-url>

git push -u origin master

```
archana@ARCHANA: /mnt/c/Users/archa/Desktop$ cd Git_Tasks/
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks$ mkdir Git_Practice
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks$ cd Git_Practice/
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ touch code.txt
touch log.txt
touch output.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ ls
code.txt  log.txt  output.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice/.git/
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ git add code.txt output.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   code.txt
        new file:   output.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        log.txt

archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ git commit -m "Added code.txt and output.txt files"
[master (root-commit) 0f8e988] Added code.txt and output.txt files
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 code.txt
 create mode 100644 output.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ https://gitlab.com/gsarchana93/git-practice.git
-bash: https://gitlab.com/gsarchana93/git-practice.git: No such file or directory
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ git remote add origin https://gitlab.com/gsarchana93/git-practice.git
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ git remote -v
origin  https://gitlab.com/gsarchana93/git-practice.git (fetch)
origin  https://gitlab.com/gsarchana93/git-practice.git (push)
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ git branch -M main
git push -u origin main
Username for 'https://gitlab.com': gsarchana93
Password for 'https://gsarchana93@gitlab.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 240 bytes | 8.00 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: To create a merge request for main, visit:
remote:   https://gitlab.com/gsarchana93/git-practice/-/merge_requests/new?merge_request%5Bsource_branch%5D=main
remote:
To https://gitlab.com/gsarchana93/git-practice.git
 * [new branch]    main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_Practice$ |
```

Question 2

Create feature1.txt & feature2.txt in master.
Create branches develop, feature1, feature2.
Stash develop.txt.
Commit new.txt in feature1.
Unstash in develop and commit.

Solution - Commands Used

```
git init
touch feature1.txt feature2.txt
git add .
git commit -m "Initial commit"
```

```
git branch develop
git branch feature1
git branch feature2
```

```
git checkout develop
touch develop.txt
git stash
```

```
git checkout feature1
touch new.txt
git add new.txt
git commit -m "Added new.txt"
```

```
git checkout develop
git stash pop
git add develop.txt
git commit -m "Committed develop.txt after unstash"
```

```
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks$ mkdir Git_tasks
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks$ cd Git_tasks/
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ ls
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks/.git/
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git branch
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ touch feature1.txt
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ touch feature2.txt
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git add .
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git commit -m "adding new files"
[master (root-commit) 9e5b569] adding new files
2 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 feature1.txt
create mode 100644 feature2.txt
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git branch
* master
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git branch develop
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git branch feature1
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git branch feature2
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git switch develop
Switched to branch 'develop'
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ ls
feature1.txt  feature2.txt
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ touch new.txt
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ stash -u
Command 'stash' not found, did you mean:
  command 'sash' from deb sash (3.8-5)
  command 'kstash' from deb heimdal-kdc (7.7.0+dfsg-3ubuntu1)
  command 'smash' from deb node-smash (0.0.15-3)
Try: sudo apt install <deb name>
archana@ARCHANA: /mnt/c/Users/archana/Desktop/Git_Tasks/Git_tasks$ git stash -u
```

```

archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git stash -u
Saved working directory and index state WIP on develop: 9e5b569 adding new files
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git switch feature1
Switched to branch 'feature1'
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ touch new1.txt
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git add .
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git commit -m "adding new file"
[feature1 146b7cc] adding new file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 new1.txt
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git switch develop
Switched to branch 'develop'
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git unstash pop
git: 'unstash' is not a git command. See 'git --help'.
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git stash pop
Already up to date.
On branch develop
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    new.txt

nothing added to commit but untracked files present (use "git add" to track)
Dropped refs/stash@{0} (8a3add9b6c59b77c78d75fda3908dfe9315f870b)
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git add new.txt
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git commit -m "adding unstashed file"
[develop 7d8789d] adding unstashed file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 new.txt
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git status
On branch develop
nothing to commit, working tree clean

```

Question 3

Create branches develop, f1, f2.

Commit main.txt in master.

Push all branches.

Delete f2 locally and remotely.

Solution - Commands Used

git init

touch main.txt

git add main.txt

git commit -m "Added main.txt"

git branch develop

git branch f1

git branch f2

git checkout develop

touch develop.txt

git add develop.txt

git commit -m "Added develop.txt"

git checkout f1

touch f1.txt

git add f1.txt

git commit -m "Added f1.txt"

git checkout f2

touch f2.txt

git add f2.txt

git commit -m "Added f2.txt"

git push origin --all

git branch -d f2

git push origin --delete f2

```
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ mkdir git_project
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ cd git_project/
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project/.git/
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch main.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "Added main.txt in master branch"
[master (root-commit) 23c238d] Added main.txt in master branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 main.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch
* master
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch develop
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch f1
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch f2
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch develop
Switched to branch 'develop'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch develop.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "Added develop.txt in develop branch"
[develop e82d3d3] Added develop.txt in develop branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 develop.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch f1
Switched to branch 'f1'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch f2.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "Added f2.txt in f2 branch"
[f1 90b599d] Added f2.txt in f2 branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 f2.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch f1
Already on 'f1'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch f1.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "Added f1.txt in f1 branch"
[f1 35a7c6d] Added f1.txt in f1 branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 f1.txt
```

```
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch f1
Switched to branch 'f1'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch f2.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "Added f2.txt in f2 branch"
[f1 90b599d] Added f2.txt in f2 branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 f2.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch f1
Already on 'f1'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch f1.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "Added f1.txt in f1 branch"
[f1 35a7c6d] Added f1.txt in f1 branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 f1.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git remote -v
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git remote add origin https://github.com/Archana-GS/Linux-Workbook.git
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git remote -v
origin https://github.com/Archana-GS/Linux-Workbook.git (fetch)
origin https://github.com/Archana-GS/Linux-Workbook.git (push)
```

```
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git push -u origin master --force
Username for 'https://github.com': Archana-GS
Password for 'https://Archana-GS@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 228 bytes | 12.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Archana-GS/Linux-Workbook.git
 * 23c238d master -> master (forced update)
Branch 'master' set up to track remote branch 'master' from 'origin'.
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git push -u origin develop
Username for 'https://github.com': Archana-GS
Password for 'https://Archana-GS@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 260 bytes | 23.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/Archana-GS/Linux-Workbook/pull/new/develop
remote:
To https://github.com/Archana-GS/Linux-Workbook.git
 * [new branch] develop -> develop
Branch 'develop' set up to track remote branch 'develop' from 'origin'.
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git push -u origin f1
Username for 'https://github.com': Archana-GS
Password for 'https://Archana-GS@github.com':
Enumerating objects: 76, done.
Counting objects: 100% (76/76), done.
Delta compression using up to 16 threads
Compressing objects: 100% (74/74), done.
Writing objects: 100% (75/75), 1.13 MiB | 1.01 MiB/s, done.
Total 75 (delta 17), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (17/17), done.
remote:
remote: Create a pull request for 'f1' on GitHub by visiting:
remote:   https://github.com/Archana-GS/Linux-Workbook/pull/new/f1
remote:
To https://github.com/Archana-GS/Linux-Workbook.git
 * [new branch] f1 -> f1
Branch 'f1' set up to track remote branch 'f1' from 'origin'.
```

```
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git push -u origin f2
Username for 'https://github.com': Archana-GS
Password for 'https://Archana-GS@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'f2' on GitHub by visiting:
remote:   https://github.com/Archana-GS/Linux-Workbook/pull/new/f2
remote:
To https://github.com/Archana-GS/Linux-Workbook.git
 * [new branch] f2 -> f2
Branch 'f2' set up to track remote branch 'f2' from 'origin'.
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch -d f2
Deleted branch f2 (was 23c238d).
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git push origin --delete f2
Username for 'https://github.com': Archana-GS
Password for 'https://Archana-GS@github.com':
To https://github.com/Archana-GS/Linux-Workbook.git
- [deleted] f2
```

Question 4

Branch creation, merging public branches, updating private branch.

Solution - Commands Used

```
touch master.txt
```

```
git add master.txt
```

```
git commit -m "Added master.txt"
```

```
git branch public1
```

```
git branch public2
```

```
git branch private
```

```
git checkout public1
```

```
touch public1.txt
```

```
git add public1.txt
```

```
git commit -m "Added public1.txt"
```

```
git checkout master
```

```
git merge public1
```

```
git merge public2
```

```
git checkout private
```

```
echo "Updated content" >> master.txt
```

```
git add master.txt
```

```
git commit -m "Updated master.txt"
```

```
git checkout public1
```

```
git merge private
```

```
git checkout public2
```

```
git merge private
```

```
git checkout master
```

```
git merge private
```

```
git checkout private
```

```
git merge master
```

```

archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch master.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "added master.txt in master"
[master d192293] added master.txt in master
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100604 master.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch
* master
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch public1
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch public2
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git branch private
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch public1
Switched to branch 'public1'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ touch public1.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "added public1.txt in public1 branch"
[public1 7792d85] added public1.txt in public1 branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100604 public1.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
(Use "git push" to publish your local commits)
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git merge public1
Updating d192293..7792d85
Fast-forward
 public1.txt | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100604 public1.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git merge public2
Already up to date.
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ ls
main.txt  master.txt  public1.txt

```

```

archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch private
Switched to branch 'private'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ vi master.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git add master.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git commit -m "Updated master.txt in private branch"
[private 4eb437c] Updated master.txt in private branch
1 file changed, 1 insertion(+)
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch public1
Switched to branch 'public1'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git merge private
Merge made by the 'ort' strategy.
 master.txt | 1
1 file changed, 1 insertion(+)
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch public2
Switched to branch 'public2'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git merge private
Updating d192293..4eb437c
Fast-forward
 master.txt | 1
1 file changed, 1 insertion(+)
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 2 commits.
(Use "git push" to publish your local commits)
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git merge private
Merge made by the 'ort' strategy.
 master.txt | 1
1 file changed, 1 insertion(+)
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git switch private
Switched to branch 'private'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git_project$ git merge master
Updating 4eb437c..7cf8a3c
Fast-forward
 public1.txt | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100604 public1.txt

```

Question 5

Implement Git Flow with feature and hotfix branches.

Solution - Commands Used

```

git init
git checkout -b develop
git checkout -b feature/login

```

```

git checkout develop
git merge feature/login

```

```

git checkout master
git merge develop

```

```

git checkout -b hotfix/urgent master
echo "Urgent fix" > urgent.txt
git add urgent.txt
git commit -m "Hotfix added"

```

git checkout master
git merge hotfix/urgent

git checkout develop
git merge hotfix/urgent

```
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ mkdir gitflow_project
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ cd gitflow_project/
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project/.git/
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ touch master.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git add .
[master (root-commit) 8b365a6] adding master.txt in master branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 master.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git checkout -b develop
Switched to a new branch 'develop'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git branch
* develop
  master
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git checkout -b feature/login
Switched to a new branch 'feature/login'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git branch
* feature/login
  develop
  master
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ touch feature.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git add .
[feature/login d221c8a] adding feature.txt in feature branch
1 file changed, 0 insertions(+), 0 deletions(-)
Create mode 100644 feature.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git checkout develop
Switched to branch 'develop'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git merge feature/login
Updating 8b365a6..d221c8a
Fast-forward
 feature.txt | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 feature.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git checkout master
Switched to branch 'master'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git merge develop
Updating 8b365a6..d221c8a
Fast-forward
 feature.txt | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 feature.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git checkout -b hotfix/urgent
Switched to a new branch 'hotfix/urgent'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ touch urgent.txt
```

```
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git add .
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git commit -m "hotfix"
[hotfix/urgent 00673b4] hotfix
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 urgent.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git checkout master
Switched to branch 'master'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git merge hotfix/urgent
Updating d221c8a..00673b4
Fast-forward
 urgent.txt | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 urgent.txt
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git checkout develop
Switched to branch 'develop'
archana@ARCHANA: /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitflow_project$ git merge hotfix/urgent
Updating d221c8a..00673b4
Fast-forward
 urgent.txt | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 urgent.txt
```

Question 6

Initialize repository and configure .gitignore to ignore secrets.txt and *.png

Solution - Commands Used

```
mkdir gitignore_project
cd gitignore_project
git init
```

```
touch notes.txt secrets.txt image.png
```

```
echo "secrets.txt" > .gitignore
echo "*.png" >> .gitignore
```

```
git status
```

```

archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ ls
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ cd gitignore_project/
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project/.git/
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project$ touch notes.txt
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project$ touch secrets.txt
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project$ touch image.png
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project$ touch .gitignore
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project$ vi .gitignore
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/gitignore_project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)

```

Question 7

Set up SSH authentication with GitHub and push without password.

Solution - Commands Used

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

```
eval "$(ssh-agent -s)"
```

```
ssh-add ~/.ssh/id_ed25519
```

```
cat ~/.ssh/id_ed25519.pub
```

Add key in GitHub -> Settings -> SSH Keys

```
ssh -T git@github.com
```

```
git clone git@github.com:username/repository.git
```

```
cd repository
```

```
echo "Testing SSH" >> test.txt
```

```
git add test.txt
```

```
git commit -m "SSH test"
```

```
git push origin master
```

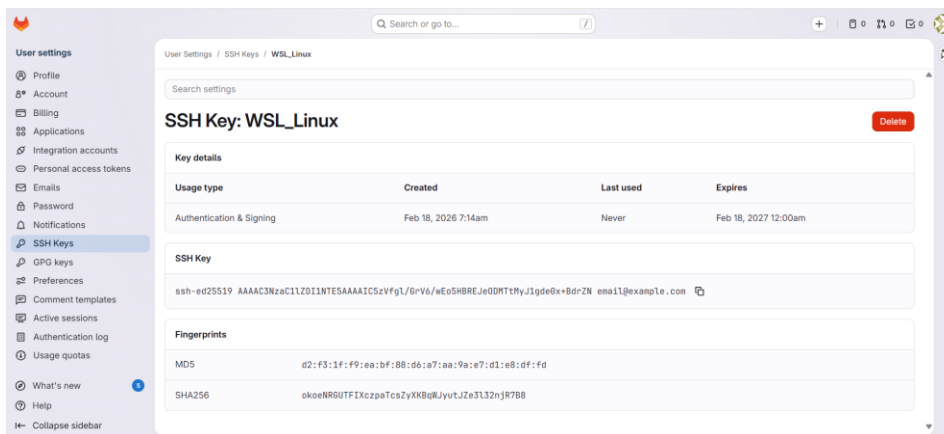


```

archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ ssh-keygen -t ed25519 -C "email@example.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/archana/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/archana/.ssh/id_ed25519
Your public key has been saved in /home/archana/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:okoeNRGUTFIXczpaTcsZyXKBqWJyutJZe3l32njR7B8 email@example.com
The key's randomart image is:
+--[ED25519 256]--+
|      .+==o      |
|      .oo+Bo++   |
|      ..+o=      |
|      .+ .+ .    |
|      = .+ . S   o |
|      . .o . .   o |
|      ooo . . .   o E|
|      oooo . o . .oo .|
|      .o . .o+ . .  |
|      o      +-----+ [SHA256]
+-----+

archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ eval "$(ssh-agent -s)"
Agent pid 58904
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ ssh-add ~/.ssh/id_ed25519
Identity added: /home/archana/.ssh/id_ed25519 (email@example.com)
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIC5zVfgl/GrV6/wEoS5HBREJeODMTtMyJlge0x+BdrZN email@example.com
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ ssh -T git@gitlab.com
The authenticity of host 'gitlab.com (172.65.251.78)' can't be established.
ED25519 key fingerprint is SHA256:eUXGm1YGsMAS7vkcx6JOJdOGHPem5gQp4taicFCLB8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'gitlab.com' (ED25519) to the list of known hosts.
Welcome to GitLab, @gsarchana93!
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ git clone git@gitlab.com:gsarchana93/git-practice.git
Cloning into 'git-practice'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (0/0), done.
remote: Total 15 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), done.

```



```

archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ ls
git-practice
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks$ cd git-practice/
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git-practice$ ls
add_calculator
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git-practice$ vi add_calculator
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git-practice$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   add_calculator

no changes added to commit (use "git add" and/or "git commit -a")
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git-practice$ git add .
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git-practice$ git commit -m "added multiplication function"
[master b5242ea] added multiplication function
1 file changed, 2 insertions(+), 1 deletion(-)
archana@ARCHANA:/mnt/c/Users/archa/Desktop/Git_Tasks/Git_tasks/git-practice$ git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 281 bytes | 31.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To gitlab.com:gsarchana93/git-practice.git
ba57e80..b5242ea  master -> master

```